Darshan Kakkad, Ph.D.

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Professional History

Senior Lecturer (Assistant Prof.), University of Hertfordshire, UK — *Jan 2025 – present*
CAR Fellow, University of Hertfordshire, UK — *July 2024* – Jan 2025
STScI fellow, Space Telescope Science Institute, USA — *2022–2024*
ESO fellow, European Southern Observatory, Chile & University of Oxford, UK — *2017–2021*

Education

Ph.D. in Astronomy, Ludwig Maximilian University, Germany — *2014–2017*
M.Sc. in Physics, Indian Institute of Technology-Madras, India — *2014*
B.Sc. (Hons) in Physics, University of Delhi, India — *2012*

Awards and Grants

Grant to organize RAS Specialist discussion on AGN outflows — 2025
Grant to organise STScI spring symposium (co-chair) — 2024
JWST General Observer grant, Program GO-03257, NASA (*PI, ~$350,000*) — *2023*
JWST General Observer grant, Program GO-03158, NASA (*co-I, ~$150,000*) — *2023*
Director's research grant, STScI (*co-PI, 1-year student grant*) — *2023*
Space Telescope Science Institute Fellowship *(~$250,000*) — *2022–2026*
Space and Astronomy Summer Program, STScI Baltimore (*Intern student grant*) — *2022*
ESO-studentship grant for a PhD student — *2020–2022*
European Southern Observatory Fellowship (*~€220,000*) — *2017–2021*
Science Support Discretionary Fund, European Southern Observatory *(€25,000*) — *2018–2020*

- Short-term grants for B.Sc. and M.Sc. thesis projects.

International Max Planck Research Scholar **(***€65,000***)** — *2014–2017*
All India 4th rank in lectureship eligibility test (Physics), UGC, India — *2014*
DAAD New Passage to India Scholarship (*€4,100*) — *2013*

Selected Observing proposals (as PI & co-I)

PI, 97 hours, VLT/MUSE — *2025*
- A MUSE filler programme to study supermassive black hole growth and feedback

PI, 13 hours, WHT/WEAVE — *2025*
- WEAVE-IFU follow-up of featureless dwarf galaxies in COSMOS

PI, 6 hours, VLT/MUSE — *2024*
- Resolving parsec-scale outflows in nearby X-ray AGN with MUSE-NFM

PI, ½ night, Keck/OSIRIS — *2024*
- Jet-ISM interaction: Warm molecular and hot ionized gas kinematics in nearby AGN with OSIRIS

PI, 29 hours, JWST/MIRI — *2023*
- Systematic search for warm H_2 gas in AGN and star forming galaxies at z=2 with MIRI

PI, 36 hours, VLT/MUSE — *2022-2023*
- Resolving gas kinematics and stellar populations in low luminosity AGN.

PI, 38 hours, VLT/MUSE — *2020-2022*
- Unveiling the jet-ISM interaction in low redshift AGN host galaxies.

Co-I, 4MOST programme: Chilean AGN/Galaxy Evolution Survey (CHANGES) — *2022*

- >1,000,000h of fibre time as a part of 4MOST community survey.

Co-I, 9 hours, JWST/MIRI *2021*
- A missing piece of puzzle: the warm molecular phase of AGN driven outflows at cosmic noon.

Co-I, 25 hours, ALMA *2021*
- AGN-galaxy interplay through multi-phase outflows and feedback.

PI, 36 hours, JVLA *2020*
- Relation between radio properties and ionized gas outflows in high redshift AGN.

Co-I, 280 hours, VLT/SINFONI *2016-2019*
- SINFONI survey for unveiling the Physics and Effect of Radiative Feedback.

PI, 4 hours, VLT/MUSE *2018*
- The interplay between inflow and outflow in a nearby merger.

PI, 20 hours, VLT/XSHOOTER *2017*
- Near-infrared spectroscopic follow-up of high-z AGN to unveil the impact of negative feedback

>35 successful proposals as co-I, equivalent to ~600 hours of X-ray, optical, near-infrared, sub-mm imaging and spectroscopy (JWST/MIRI, VLT/SINFONI, XSHOOTER, APEX, ALMA, ALMA-ACA, NICER).

Service

Leadership

Astro thematic panel member, UK Space Frontiers 2035, UKSA *2025–2026*
Deputy equality committee chair, University of Hertfordshire, UK *2025–present*
Co-lead, Widefield Spectroscopic Telescope AGN Science group 2023–present
Postdoc Representative, Space Telescope Science Institute, USA *2022–2024*

Conference & seminar organization

Co-Chair, EAS-2026 Special Session, Lausanne, Switzerland *June 2026*
Co-Chair, RAS Specialist discussion on AGN-driven outflows *October 2025*
Co-Chair, STScI Spring Symposium *April 2024*
SOC member, National Astronomy Meeting, Birmingham, UK *July 2026*
SOC member, AGN-FAAST meeting, ESO-Garching, Germany *July 2026*
Organizer, CAR seminars, University of Hertfordshire *2024–2025*
Organizer, Galaxies/AGN Journal club, STScI *2023–2024*
Organizer, HotSci talk series, STScI *2023*
Organizer, Monthly lecture series, ESO-Chile *2019–2020*
LOC member, Torus 2018, Puerto Varas, Chile *Dec 2018*

Referee for high-impact peer-reviewed journals
- Monthly Notices of the Royal Astronomical Society, UK *2019–present*
- Astronomy & Astrophysics, EU *2020–present*
- The Astrophysical Journal, US *2020–present*

4MOST, Project Culture Working group *2023–present*
SDSS-V, *representative* of the Committee on Inclusiveness *2023–2024*
User Support for COS instrument, Hubble Space Telescope *2022–2024*
Leveler, JWST Cycle-2 Telescope Allocation Committee *2023*
Selection panel member, Summer program for undergraduates, STScI *2022–2023*
Thesis committee panel member, Pontifica Universidad de Chile *2021*
Selection panel member, ESO-fellowship selection committee *2018–2020*
Technical review panel member, ESO Director’s Discretionary Time *2017–2020*
Member of MUSE instrument operations team & UT4/VLT team *2017–2020*
Selection panel member, ESO-studentship selection committee *2016–2020*
Scientific assistant, Observing Programs Office, ESO *2016–2017*

Major projects & international collaborations

COSMOS-3D collaboration 2025 - present
COSMOS-Web collaboration 2024 – present
GOALS survey 2024–present
QFeedS survey 2022 - present
4MOST Chilean AGN/Galaxy Evolution Survey (ChANGES) *2022–present*
SUPER survey (www.super-survey.org) *2015–present*

- Managing current and future data release publications.

BASS survey (www.bass-survey.com) *2018–present*

- Leading the MUSE follow-up of X-ray AGN, primary advisor of a Ph.D. student.

Recent seminars, talks and lectures

25+ invited talks and seminars at international conferences and research institutes and 40+ contributed talks and posters at conferences. Selected few talks are highlighted here:

Invited speaker, The multiscale environment of AGN across cosmic time, Germany *Nov 2025*
Invited review speaker, EAS 2025, Ireland *June 2025*
Invited Panel member (Careers), Bridge Irena, York, UK *June 2025*
Invited colloquium, *Oxford University, UK* *Feb 2025*
Invited seminar, *Newcastle University, UK* *Oct 2024*
Invited seminar, NASA-Goddard Space Flight Centre, Maryland, US *May 2024*
Invited seminar, Institute of Astrophysics, Canary Islands, Spain, US *Oct 2023*
Invited seminar, University of Hertfordshire, UK *Sept 2023*
Invited seminar, University of Cambridge, UK *April 2023*
Astronomy for non-Astronomers, STScI, US *April 2023*
Outreach talk, Oxford, MD, US *March 2023*
Invited Seminar, University of Lancaster, UK *Feb 2023*
Public lecture, STScI, US *Feb 2023*

- https://www.youtube.com/watch?v=8a6PrAk84eM

Invited colloquium, UCM, Spain *Nov 2022*
Invited seminar, University of Southampton, UK *Mar 2022*
Invited seminar, STScI-Baltimore, US *Sept 2021*
Invited seminar, University of Sau Paolo, Brazil *June 2021*
Invited talk, DARK, Copenhagen, Denmark *June 2019*
Invited seminar, AGN meeting, Santiago, Chile *May 2019*
Invited talk, AGN club, Leiden Observatory *Oct 2018*

Teaching & Supervision

Lecturer

Machine Learning & Neural Networks, M.Sc. Data Science, U. Hertfordshire *2026–present*
Data Science Projects, M.Sc. Data Science, U. Hertfordshire *2025–2026*
AGN Physics, *Invited guest lecturer*, Andres Bello National University, Chile *2019–2020*

Tutor

Linear Modelling (R programming), B.Sc. Mathematics, U. Hertfordshire *2024–2025*
Data Science Projects (supervisor), M.Sc. Data Science, U. Hertfordshire *2024–present*

Supervision

Co-advisor for 6 students and a primary advisor of five students. Details available upon request.

Technical Skills

- Programming & software experience: Python, IDL, CASA, Latex documentation, MS office (advanced), C, C++, IRAF, Bash, HTML, R & ZEMAX OpticStudio (basic)
- Experienced with instrument commissioning and science operations on ground-based (Very Large Telescope, VLT) & space telescopes (Hubble Space Telescope).
- >180 nights observing, quality checks, data reduction and analysis experience at the VLT with imaging and spectroscopic instruments: FLAMES, UVES, XSHOOTER, SINFONI, MUSE, HAWK-I, VISIR, KMOS.

Publications (as first/second author)

1. **D. Kakkad**, Y. Song, T. S. Y. Lai et al. (submitted); GOALS-JWST: Resolved multi-phase molecular gas in IRAS 20551-4250 using JWST and ALMA, MNRAS
2. **D. Kakkad**, I. Lazar, S. Harish et al. (submitted); JWST observations of dwarf galaxies in COSMOS-Web: the ISM properties of dwarfs in the nearby Universe, MNRAS
3. **D. Kakkad**, V. Mainieri, T. S. Tanaka et al. (2025); JWST MIRI/MRS observations of hot molecular gas in an AGN host galaxy at Cosmic Noon, MNRAS, in press
4. **AGN co-lead** of The Wide-field Spectroscopic Telescope (WST) Science white paper, Mainieri et al. (2024), wrote Section 5.7 of the white paper article; arxiv:2403.05398
5. **D. Kakkad**, M. Stalevski, M. Kashimoto et al. (2023a); Dissecting the active galactic nucleus in Circinus IV. MUSE NFM observations unveil a tuning-fork ionised outflow morphology, MNRAS, 519, 5324
6. **D. Kakkad**, V. Mainieri, G. Vietri et al. (2023b); SUPER VII. Morphology and kinematics of Hα emission in AGN host galaxies at Cosmic noon using SINFONI, MNRAS, 519, 5324
7. **D. Kakkad,** N. Thatte, M. Tecza, et al. (2022b); Characterisation of Line Spread Function in Integral Field Spectrographs from ground-based telescopes, Proc. SPIE 12184, 1218460,
8. **D. Kakkad**, E. Sani, N. Mallmann et al. (2022a); BASS XXXII: Outflow scaling relations in low redshift X-ray AGN host galaxies with MUSE, MNRAS, 511, 2
9. **D. Kakkad**, M. Tecza, N. Thatte et al. (2020b); HARMONI: Characterizing the line spread function with a tunable Fabry-Pérot etalon, Proc. SPIE 11451, 114515W
10. **D. Kakkad**, V. Mainieri, G. Vietri et al. (2020a); SUPER II: Spatially resolved ionized gas kinematics and scaling relations in z~2 AGN host galaxies, A&A, 642, A147
11. J. Hartke, **D. Kakkad**, et al. (2020); MUSE+GALACSI: The first years; Proc. SPIE 11448, 114480V
12. **D. Kakkad**, B. Groves, M. Dopita et al. (2018); Spatially resolved electron density in the narrow line region of z < 0.02 radio AGNs, A&A, 618, A6
13. **D. Kakkad**, V. Mainieri, M. Brusa et al. (2017); ALMA observations of cold molecular gas in AGN hosts at z~1.5—Evidence of AGN feedback?, MNRAS, 468, 4
14. **D. Kakkad**, V. Mainieri, P. Padovani et al. (2016); Tracing outflows in AGN forbidden region with SINFONI, A&A, 592, A148

15. D. Kakkad, N. Indriolo, T. Ake et al. (2025); A revised geometric distortion correction for the far-ultraviolet detector of the Cosmic Origins Spectrograph, cos rept., 10
16. **D. Kakkad**, E. Frazer & K. Rowlands (2023c); Changes in the background locations for COS/NUV XTRACTAB, cos rept.

Publications (as a contributing author)

All publications available here:
https://ui.adsabs.harvard.edu/search/fq=%7B!type%3Daqp%20v%3D%24fq_database%7D&fq_database=(database%3Aastronomy%20OR%20database%3Aphysics)&q=%20author%3A%22Kakkad%2C%20D.%22&sort=date%20desc%2C%20bibcode%20desc&p_=0